

C Programming

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Agenda

- Tokens
- Keywords
- Identifiers
- Data Types, Variables & Constants
- Data Types
- Format Specifier
- sizeof operator
- Printf()
- Escape Sequence



Tokens

- C program is **made up of functions**.
- Function is made up of **statements**.
- Statement contain **multiple tokens**.
 - Keywords
 - Data Types
 - Identifiers
 - Variables
 - Constants
 - Operators



Keywords

- Keywords are **predefined words** used in program, which have **special meanings** to the compiler.
- They are **reserved words**, so **cannot be used as identifier**.
- K & R C has **27 keywords**. C89 added 5 keywords. C99 added **5 new keywords**

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



Identifiers

- Identifiers give **names to variables, functions, defined types and pre-processor** macros.
- **Rules of Identifiers:**
 - Should **start** with **alphabet or with _** (underscore)
 - Can include alphabets, _ (underscore), digits
 - Case sensitive (upper and lower case letters are different)
- **Examples:**
 - Var_1 //Valid
 - 1_var // Not Valid
 - _var //valid
 - Var-1 // invalid
 - Basic Salary //invalid



Data Types, Variables & Constants

- C allows computations to be performed on **various types of data**.
 - **Numerical: Whole numbers, Real numbers**
 - **Character: Single character, Strings**
- **Fixed data values** are said to be **constants**.
 - 12, -45, 0, 2.3, 76.9, 1.23456e+2, 'A', "Sunbeam", etc.
- Data is **hold in memory** locations identified by names called as **variables**.
 - Variable must be declared before its use in the program.
 - As per need, variable have some data type.
- Simple C data types are: **int, double, char**.
 - Data type represents **amount of space assigned** to the variable.
 - It also defines internal storage of the **data**.



Data Types, Variables & Constants

- **Variable examples**

- `int number = 10;`
- `double basic_salary = 20000.0;`
- `char letter = 'A';`
- `int roll_number;`
- `roll_number = 20;`
- `double price = 200.0;`
- `price = 300.0;`
- Each variable is assigned some memory location.
- Size of data type of given variable or constant is found by `sizeof()` operator.



Data Types

- Data types in C refer to an extensive system used for declaring variables or functions of different types.
- The type of a variable determines how much **space it occupies in storage**.
- **Basic data types (primitive)**
 - 1.character(char)
 - 2.Integer(int)
 - 3.Single precision floating point (float)
 - 4.Double precision floating point(double)
 - 5.void
- **Derived data types : Derived from basic data types**
 - 1.Array type
 - 2.Pointer type
 - 3.Function type

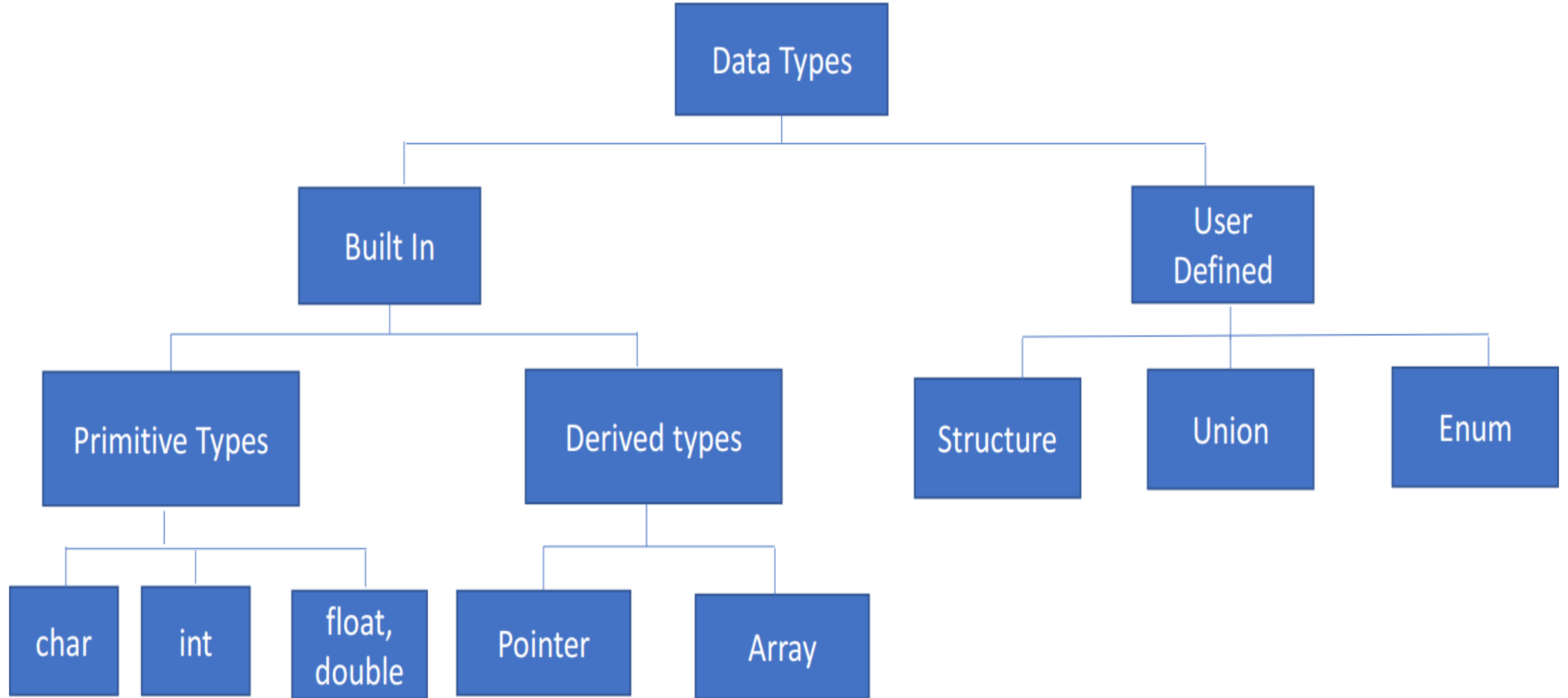


Data Types

- User defined data types
 - 1.Structure
 - 2.Union
 - 3.Enumeration



Data Types



Data Types

C Basic Data Types	32-bit CPU		64-bit CPU	
	Size (bytes)	Range	Size (bytes)	Range
char	1	-128 to 127	1	-128 to 127
short	2	-32,768 to 32,767	2	-32,768 to 32,767
int	4	-2,147,483,648 to 2,147,483,647	4	-2,147,483,648 to 2,147,483,647
long	4	-2,147,483,648 to 2,147,483,647	8	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
long long	8	9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	8	9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4	3.4E +/- 38	4	3.4E +/- 38
double	8	1.7E +/- 308	8	1.7E +/- 308



ASCII Value

- **ASCII (American Standard Code for Information Interchange)** is a standard character encoding used in telecommunication.
- ASCII codes are used to represent alphanumeric data.
- it currently defines 95 printable characters including **26 upper case letters (A to Z)**, **26 lower case letters**, **10 numerals (0 to 9)**, and **33 special characters** including **mathematical symbols**, **punctuation marks**, and **space characters**.



ASCII Value Chart

Dec	Hex	Oct	Chr	Dec	Hex	Oct	HTML	Chr	Dec	Hex	Oct	HTML	Chr	Dec	Hex	Oct	HTML	Chr
0	0	000	NULL	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	Start of Header	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	Start of Text	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	End of Text	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	End of Transmission	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	Enquiry	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	Acknowledgment	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	Bell	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	Backspace	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	Horizontal Tab	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	Line feed	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	Vertical Tab	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	Form feed	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	Carriage return	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	Shift Out	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	Shift In	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	Data Link Escape	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	Device Control 1	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	Device Control 2	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	Device Control 3	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	Device Control 4	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	Negative Ack.	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	Synchronous idle	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	End of Trans. Block	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	Cancel	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	End of Medium	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	Substitute	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	Escape	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	File Separator	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	Group Separator	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	Record Separator	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	Unit Separator	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		Del

sizeof operator

- sizeof is a **compile time operator**.
- It is used to **calculate the size of its operand**.
- It **returns the size(how much memory required)** of the variable **in bytes**.
- When sizeof is applied with datatypes, it simply return the amount of memory allocated to that datatype.
- `Int num = 10;`
- `Printf("%d",sizeof(num));`
- Output is 4. as the size of integer datatype is 4 bytes
- When sizeof is applied with expressions it returns the size of the expression.
- `Char ch = 'A';`
- `printf("%d",sizeof(ch)); // returns 1`
- `printf("%d",sizeof('A')); // returns 4`
- `printf("%d",sizeof(num+ch)); // returns 4`



Printf()

- printf is a **standard library function** used to write formatted output to the standard output device (usually the screen)
- Requires **header file `stdio.h`**
- Format **`printf( formatstring [, value ] ... );`**
- Should have one **format specifier** for **each value listed**, and types should match
- Example - `printf("Value = %.2f",temp);`
- Arbitrary strings and variable values can be printed using `printf()` function.
- Int - `%d`
- double - `%lf`
- char - `%c`
- ...
- Examples:
 - `printf("Hello PreCAT @ Sunbeam");`
 - `printf("%d", roll_number);`
 - `printf("%d %lf %c", number, basic_salary, letter);`
 - `printf("Book price is %lf", price);`



Escape Sequence

- \n - New line
Moves cursor to the beginning of next line
- \t - Horizontal tab
Moves the cursor to next horizontal tab stop
- \r - Carriage return
Moves the cursor to the beginning of current line
- \b – Backspace
Moves cursor one position to the left of its current position
- (Note: Its effect cant be visualized on eclipse , it can be visualized when we use printer)
- \f - Form Feed
Moves cursor to the beginning of next page
- \v - Vertical tab
Vertical tab
- \a - Alert
Alerts by Generating beep
- \' - Single quote
mark prints '
- \" - Prints
Double quote mark prints"
- \\ - Backslash Character
Prints\



Thank You

